Sparnaay, M. J.
1958

Physica XXIV
751–764

MEASUREMENTS OF ATTRACTIVE FORCES BETWEEN FLAT PLATES

by M. J. SPARNAAY

Philips Research Laboratories, N.V. Philips' Gloeilampenfabrieken, Eindhoven, Nederland

Abstract

Results of experiments concerning attractive forces between flat metal plates, and a description of the apparatus used, are given. The observed attractions do not contradict Casimir's theoretical prediction.

An explanation of results of earlier measurements made by the author on the attraction between clean, and also between silvered glass and quartz plates, all brought in an ionized atmosphere, is suggested. These results were due to differences in the surface potentials of the opposing plates.

1. *Introduction.* Casimir [1]) derived the following expression for the attractive force F, which exists between two metal plates a distance d apart:

$$F = 0.013/d^4 \text{ dynes/cm}^2 \tag{1}$$

The distance d, which is in this expression measured in microns, must be larger than the penetration depth of electromagnetic waves in the metal. The penetration depth may be estimated to be of the order of 0.1 micron.

If the plates are of dielectric material, the attractive force can be expressed as:

$$F = 0.013\, f(\varepsilon)/d^4 \text{ dynes/cm}^2 \tag{2}$$

where $f(\varepsilon)$ is a complicated function of the dielectric constant ε. For large ε, the function $f(\varepsilon)$ approaches unity, for small ε, $f(\varepsilon)$ is smaller than unity. Eq. (2) follows from a theory given by Lifshitz [2]). The penetration depth in the dielectric case, and $f(\varepsilon)$ also, depend on the dispersion properties of the dielectric, a large dispersion (at low frequencies) leading to a large limiting value of d.

If d is shorter than the penetration depth, eqs (1) and (2) both reduce to the Hamaker–De Boer equation [3]), which is based on the existence of London-Van der Waals forces between the atoms in both plates.

$$F = A/6\pi d^3 \tag{3}$$

For plates of identical material the constant $A = \pi^2 q^2 \lambda$, where q is the number of atoms per cm^3 of the plate material and, according to London's

harmonic oscillator model [4]), $\lambda = \frac{4}{3}\alpha^2 h\nu$, where α is the polarizability of an atom and $h\nu$ is its ionization energy. We have written down the expression of the constant A for the case that the London-Van der Waals forces are additive. This assumption is not necessarily valid,[5]) but the value of $A/6\pi$ may be estimated to be in between 0.02 and 0.4 for most materials. (Here again d is expressed in microns and F in dynes/cm^2). When the plates have different surface potentials and when electric charges are able to come from one surface to the other, then there is an additional attractive force, given by:

$$F = 4.5 \times 10^{-5} (V_1 - V_2)^2/d^2 \qquad (4)$$

The potential difference $(V_1 - V_2)$ is given in mV and d in microns. When $V_1 - V_2 = 17$ mV the force due to this potential difference is equal to the attractive force, given by Casimir's expression (1), when $d = 1\mu$.

Attractive forces of the range indicated by the above equations are important for the understanding of flocculation phenomena in lyophobic colloids.[6]) When such a colloid system looses its stability, i.e. when it flocculates, the colloid particles stick together and form large agglomerates. When the particles are all of the same kind, the phenomenon of flocculation is often ascribed to the existence of attractive forces of the type, given by eqs. (1), (2) and (3). When there are particles of different kind present in the system, eq. (4) may also account for flocculation.[7]) Eq. (4) may also be applied to the large attraction found by Von Buzagh [8]) in 1929 between glass spheres (radius 1–10 μ) and a glass wall, both in an electrolyte solution. Glass spheres were allowed to settle on the bottom of a closed glass cell in these experiments. After reversing the cell upside down a fraction of the glass spheres did not fall off. This fraction depended on the electrolyte concentration. The experiments were confirmed by the author in 1947. Other phenomena, such as physical adsorption of gases, the very existence of non-ideal gases and so on, point to the existence of attractive forces of a general character. Direct measurements of these forces is therefore important. Such measurements are presented in this paper. The purpose was, to check eq. (1). The method used was essentially the same as that described earlier for glass and quartz plates, the main difference being, that the distance between the metal plates now used could no longer be determined by optical means. Essentially a capacity method was used instead. The determination of the distance was now less precise, but this was the only drawback of the choice of metal plates.

2. *Choice of plate material.* In general, the following requirements should be met:

1) Obstacles on the plates, larger than $\frac{1}{2}\mu$, should be removed

2) Static electric charges on the plates should be absent, or, where present, their influence should be excluded.

3) The influence of differences of the surface potentials on both plates should be eliminated.

It is difficult to meet these requirements simultaneously. In fact the second and the third requirements are mutually exclusive when the surface is non-conductive, since static electric charges can be removed by making the environment of the surface under consideration conductive, whereas potential effects do not manifest themselves when the plates are insulated.

We now discuss the arguments leading to the choice of metal plates instead of glass or quartz plates used previously.

2.1 *Obstacles.* (fig. 1*A*). Apart form dirt specks which should be removed upon a suitable mechanical cleaning procedure, atmospheric influence may give rise tot he existence of obstacles other than dirt specks. Thus most metal surfaces have an oxide layer formed by the reaction between atmospheric oxygen and the metal.

Various theories indicate that the oxide layer on chromium steel and chromium, giving rise to passivity, [10]) has 5–50 Å thickness. This small thickness, combined with the high degree of hardness of steel and chromium surfaces, and with the possibility of lapping the surface sufficiently flat (maximum protrusions less than 500 Å were obtained) made

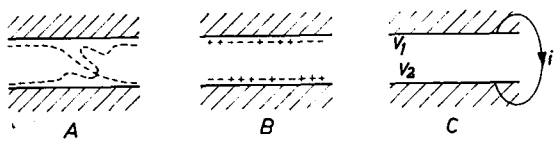


Fig. 1. Illustrations of the three factors affecting the measurements of intermolecular forces:

A) chemisorbed irregular layer (glass, quartz, possibly aluminium), decreases the attraction.

B) static electric charges, increase the attraction when situated in the given distribution.

C) different surface potentials, increase the attraction when an electric current i flows (eq. 4).

them suitable for our purpose. The oxide layer on aluminium is often thicker and the aluminium surface has a lower degree of hardness than have steel or chromium. However, when turned on a precision lathe, the flatness of an aluminium surface can be made exceedingly good, maximum protrusions less than 200 Å being present only. Therefore two aluminium plates were also used for attractive measurements. Repulsions were obtained only however and it is possible that thin sheets of the oxide layer, protruding from the surface, are responsible for this repulsion. If repulsions were found between steel plates or chromium plates, these repulsions were always due to the presence of rigid obstacles, in contradistinction with the obstacles between two aluminium plates, where the obstacles were more or less plastic and elastic in nature.

The choice of steel and chromium represents an improvement over the choice of glass or quartz used previously since chemisorption of water always gives rise to an irregularly shaped silicagel layer. Much evidence [11]) has been given previously of the existence of such a gel layer. Often repulsions were found instead of attractions at an interplate distance smaller than $1\frac{1}{2}\mu$ and in general, the gel layer has the tendency to decrease attractions. The obstacles apparently present were again plastic and elastic in nature.

It is known that a surface layer of low hardness compared with the bulk material exists: the Rayleigh-Beilby layer.[12]) It is probably the Rayleigh-Beilby layer which is beginning to form after the glass and quartz plates were polished, whose influence has been observed previously. The hygroscopicity [13]) can easily be checked after observing the continuously increasing weight of freshly dried glass ware, placed in an atmosphere which is not completely dry. The increase amounts to about 10^{-6} gram per cm^2 of the surface area after an exposure of about 15 minutes.

Also the fact, that freshly polished glass or quartz plates can easily be brought together until a distance of 200 Å is obtained whereas this is hardly possible after waiting an hour or so (followed by removal of dust particles) points to some special behaviour of the glass or quartz surfaces. It is difficult to exclude water entirely when working with polished glass or quartz plates because the polishing procedure requires water. The method which should preferably be followed to detect rapidly the presence of obstacles with a plastic and elastic nature, will be dealt with in section 3.6.

2.2 *Static electric charges.* (fig. 1*B*) It is known, that surfaces of insulating material often become charged after rubbing. Rubbing however is often required in a cleaning procedure of the surface. When the electric charges are static, i.e. when the charges don't move, imitation of eqs (2) or (3) takes place only when the charges are distributed on the surfaces as illustrated in fig. 1*B*: positive and negative charges in "mosaics", oppose each other in the opposite plates. A random distribution, and also a distribution such that the "mosaics" are large (much larger in linear dimension than the interplate distance) does not result in a force of interaction between the plates. The probability of a "mosaics" distribution giving more or less rise to imitations of eqs (2) and (3) is of course small. However in order to avoid difficulties, the surfaces must be made conductive. In the case of glass and quartz surfaces a radioactive agent was always present. Ionizations in 10^{-2} mm Hg were rigorously performed before or during the measurements. These ionizations often lasted half an hour or more. Tolansky's method [14]) of ionic bombardment was followed. Moreover, a silver layer was sometimes evaporated on the surfaces.

If the charge carriers are able to move over the surface, for example,

when they are ions in the gel layer on glass or quartz, then two surfaces, each covered with such a layer, attract each other. This may be understood[15]), when it is remembered, that a fluctuation of the charge density in a volume element in one of the surface layers induces a region of opposite charge in the other surface layer. This is precisely the same picture as that underlying the explanation of Van der Waals forces. The "mosaics" considered above form as it were a "snapshot" of this picture and that is probably the only physical meaning to be attached to them.

It was assumed, that metal surfaces were sufficiently conductive. No radio active agent was present and ionizations were not carried out.

2.3 *Different surface potentials.* (fig. 1*C*). There usually is a difference of the surface potentials on both surfaces, even when the plates are made of the same material*. An attractive force, given by eq. (4) can be found only, when charge carriers can come from one surface to the other. Thus in order to prevent potential effects, the atmosphere between the plates should not be made conductive: a radio active agent should not be present and ionizations should not be carried out. Electrical leakage through the apparatus should be eliminated. The time interval necessary for a measurement was less than a minute. Thus the time τ of discharge of the condenser formed by the two opposing plates should be much larger than a minute. We may estimate the magnitude of the resistivity R for the electric charges to come from one surface to another when we consider the product RC, where C is the capacity formed by the two plates under consideration. When the surface area of the plates is 1 cm^2 and their distance is 1μ, then $C \cong 10^3$ pF and thus in $\tau = RC$, the resistivity R should be larger than 10^9 ohm in order to make $\tau \gg 1$ minute.

Usually the difference of surface potentials between two plates of the same metal ranges from 5 to 25 mV. The value depends on the surrounding atmosphere and shows a drift, which can be of the order of 5 mV per hour.

Larger potential differences ranging between 30 mV and 500 mV were observed between two glass plates, each covered with 500 Å of gold, the gold being sputtered on the plates after cleaning them by ion bombardment. Here also a drift was present and dependence upon the surrounding atmosphere was observed. It is difficult, though not impossible, to suppress these large potential differences, one condition being special geometrical arrangements in the ionization chamber. These arrangements could not be realized in the experimental set-up used previously. The metal plates in the present apparatus were rigidly insulated. The only possibility of discharge was through the gap separating the plates. If the gap width is 1μ, then a potential difference of 100 mV gives rise already to a field strength of 100 Volt/cm. A discharge then would produce a force, which is appreciable and our next task

*) Thanks are due to Ir. Van Nie of these laboratories for communication of his results.

will be the introduction of a method for compensation of potential differences.

2.4 *Interpretation of earlier measurements.* In view of the three factors, mentioned in the previous sections, the following interpretation of the measurements of attractive forces between glass and quartz plates becomes probable: The large attractive forces, found in presence of a radio active agent and of ionization of the atmosphere, and also found when a 200 Å silver layer was present on the plates [9]), are due to a difference of the surface potentials, ranging between 100 mV and 200 mV. Thus a force which was of the order of 1–4 dyne/cm^2 at $d = 1\mu$ was often found. A decrease of this attractive force, occasionally turning into a repulsive force [11]), which was often found after introduction of water vapour or after re-ionization (or sputtering) is due to an enhanced formation of silicagel on the plates, or, in general: to damage of the surface. A decrease systematically found in an early stage of the experiments (see ref. 5, table 5, and sections 1.4 and 1.9*a* of that paper) was already ascribed to the influence of obstacles.

Such obstacles will also have affected data which were given as final results by Derjaguin *et alii* [16]) and by Kitchener and Prosser [17]). Potential effects also become probable, because the attraction between glass plates of different kind was found to be larger than the attraction between plates of the same kind [18]). Also the decay of the force upon the distance resembled more the square distance law, eq. (4), in the former case, than it did in the latter. The drift, observed in the potential measurements, was also observed in the direct attractive measurements. For example, systematic variations of a factor two (in either direction) in the course of an hour were not uncommon. These variations were absent when obstacles were present. The ultimate force distance law found previously lay in between eqs. (2) or (3), and eq. (4). Thus forces, given by eqs. (2) and (3) or also given by the fluctuation phenomena mentioned in section 2.2 may not be excluded. Especially the attractions found at 200–400 Å distance will be due to intermolecular forces, because at that distance these forces will be larger than any other force. When this distance did not obtain spontaneously upon decreasing the interplate distance, it was almost certain, that obstacles between the plates were present. The force found at this distance should be interpreted according to eq. (3), and is in agreement with the attraction, found by Howe *et alii* [19]) between two freshly molten glass surfaces at a distance of about 20 Å.

The present author overlooked the potential effect, other investigators [16) 17)] the obstacle *and* the potential effect. In view of the efforts given by these investigators to remove static electric charges, of the large attractions reported by them, for example when silicon oil was present on the plates (silicon oil will suppress gel obstacles) and in view of the fact that a distance as small as just reported was not obtained, an interpretation along the lines given here becomes probable for the small force obtained by them. These

forces (0.01–0.05 dynes/cm^2 when $d = 1\mu$) moreover considerably exceed the expectations, based on eq. (2)

3. *Experiments.* 3.1 *Apparatus.* The principles of the apparatus used were similar to those described earlier. There were two advantages: 1) The sensitivity was much increased and was now $0.1–1 \times 10^{-3}$ dyne. 2) The cleaning procedure of the metal plates was much facilitated, cleaning now being carried out when the metal plates remained fixed in the apparatus, whereas in an earlier set up removal of the plates was necessary.

A disadvantage was, that no harmonic vibrations of one of the plates, helpful for an investigation of the presence and the kind of obstacles, could be carried out (see 3.6).

The apparatus was placed in a glass cylinder whose radius was 15 cm and whose height was 30 cm. The cylinder could be closed at both sides by steel plates, the apparatus being mounted on the lower plate. Usually the system, after replacing the atmosphere by pure, dust free nitrogen gas, was evacuated until a pressure, not lower than 10^{-2} mm Hg was obtained. Then the connections between pump and apparatus were removed. The apparatus was placed on a vibration-free table, but small anharmonic vibrations were nevertheless observed. These vibrations were partly removed by making use of the viscous properties of the nitrogen gas which was left in the apparatus and which consequently also was present between the flat plates under consideration. Though in this way unexpected, relative displacements of the two plates were largely suppressed the factor vibrations was one of the factors limiting the accuracy of the measurements. Thus, at a large distance ($2\,\mu$) between the plates, the viscous influence of the dust free nitrogen was too small to prevent serious vibrations, the sensitivity then being only about 1×10^{-3} dyne.

Once the plates were assumed to be clean, the system was pumped out and all further necessary manipulations, such as bringing the plates in parallel position and displacing the plates at will, could be carried out from outside.

3.2. *Balance system.* Two types of balance systems used are shown schematically in figs. 2*A* and 2*B*. The attraction which existed between the metal plates m when the interplate distance was made sufficiently small and when obstacles were absent, gave rise to a deflection of the aluminium beam k. This deflection was measured as an increase of the capacity of the condenser c. At the left hand side of the suspension e (the suspension e consisted of a steel blade of 0.05 mm thickness, 30 mm width and the height of the part shown in the figs. was 3 mm) the copperbodies W, one at each side of e, nearly counterbalanced the weight of the upper plate m.

Equilibrium was established by the spring s whose point of suspension could be varied in vertical position as indicated in fig. 2. The weights of the

copperbodies W were varied between 0 and 20 gram and their position could be varied over a range of several cm as indicated. The spring s could be allowed to have a low constant such that the force constant of the whole balance system was made between 1 and 0,1 gram/cm. Since the capacity measurement easily allowed a determination of a distance variation of 100Å units, 10^{-7} grams could be detected.

A detection of lower force values was during these measurements prevented by the presence of anharmonic vibrations. The balance system was checked by placing thin tungsten or platinum wires of known length and cross section

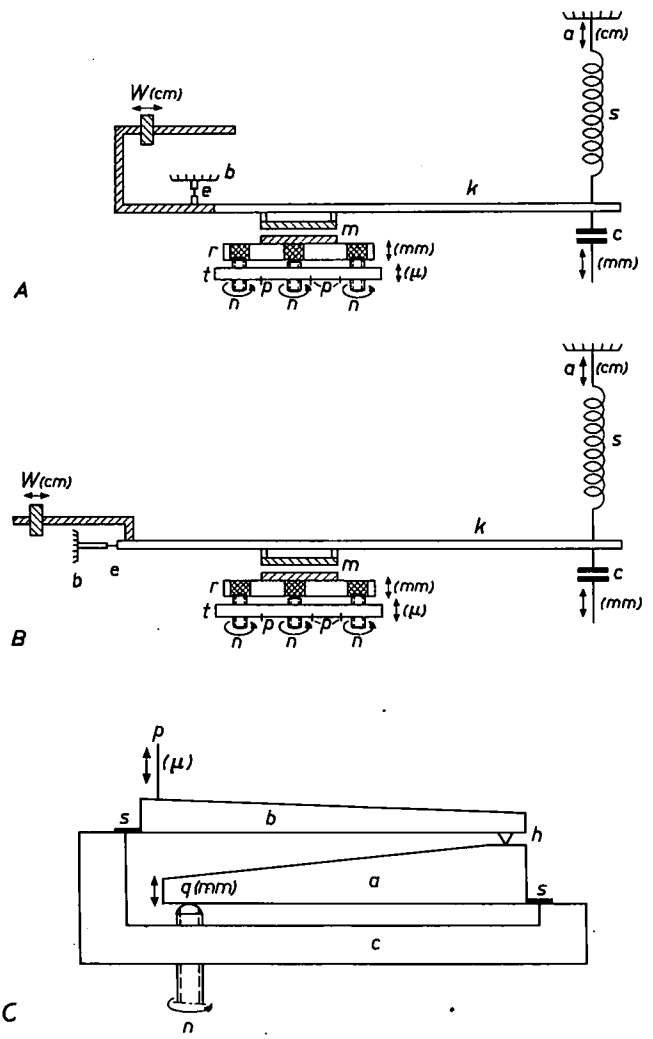


Fig. 2. *A* and *B*, two types of balances used (schematically) *C*, lever system. The arrows and the length units in brackets indicate the direction and the possible range of movements of the various parts.

on the upper plate *m* and observing the accompanying increase of the capacity *c*. Balance systems as described here are now being used for measuring gasadsorption.

3.3 *Movements of the lower plate m and the lever systems used.* The lower metal plate *m* was placed on a brass plate. This brass plate was placed on the tops of three screws *n* (0.5 mm lay) such that the plate *r* and thus also the lower plate *m* were insulated. The tangential position of the plate *r* was fixed, because one of the insulations had a grove in the bottom, one had a cone shaped gap, and the third had a flat bottom. The whole set of plates *r*, *t* and the lower plate *m* could be moved over a range of microns by means of three identical lever systems, whose position is indicated by *p* in figs. 2. These letters *p* correspond with the letter *p* in fig. 2*C* where one of the lever systems is shown, the connection consisting of a steel bar of 1mm thickness and 10 mm length. When the screw *n* was rotated 2π radians, the vertical displacement of the steel bar was 3,6 microns.

The relation between the angle of rotation of the screw *n*, and the vertical displacement of the steel bar, being of major importance for the determination of the distance between the metal plates *m*, was carefully checked. It proved to be important that the connections *s* in fig. 2*C* consisted of stiff (steel) springs. In an early stage of the experiments these connections, and also the suspension *e* (figs. 2) consisted of knife edge contacts.

Large and irreproducible hysteresis behaviour was the result, much depending on the intensity of the anharmonic vibrations of the environment, but hardly depending on the material used. Finally one knife edge contact (*h* in fig. 2C) remained, the pressure indirectly exerted by the left hand spring *s* on the steel arm *a*, over the knife edge contact, being about 200 grams. Now the hysteresis behaviour, though not completely absent, was well reproducible as is shown in fig. 3. This fig. gives the relation between the angle of rotation of screws *n*, simultaneously used in the three lever systems, and the variation of the capacity of the condenser, now formed by the two metal plates *m*, where their distance was 7.7–9.5 μ. The upper plate *m* was now fixed by moving the lower plate of the condenser *c* over such a range, that the upper plate *c* rested upon the lower plate.

The graph, representing measurements taken in about 2 hours, shows that there is some drift both in the distance between the plates and in the measurement of the capacity. The observed hysteresis may be described as follows:

A small deflection of the arm *b* relative to arm *a* is accompanied by a large force in *h* retaining the deflection. A larger deflection of the arm *b* is accompanied by a small force in *h* or no force at all. The hysteresis curves, which were the same for any possible position of the arm *a* relative to arm *b* were shown to have the general properties as given by Everett *et alii* [20]) in their theory of hysteresis phenomena:

A long delay somewhere on the curves (for example an hour) did not alter the shape of the hysteresis loops. Returning the direction of rotation of the screws n, half way the curve resulted in a "scanning curve": the hysteresis loop was crossed and the curve at the other side of the loop was followed. When the direction of rotation was inverted within $\pi/5$ radians after the beginning of measuring a curve (at the upper part of the upper curves, at the lower part of the lower curves) complete reversibility was observed.

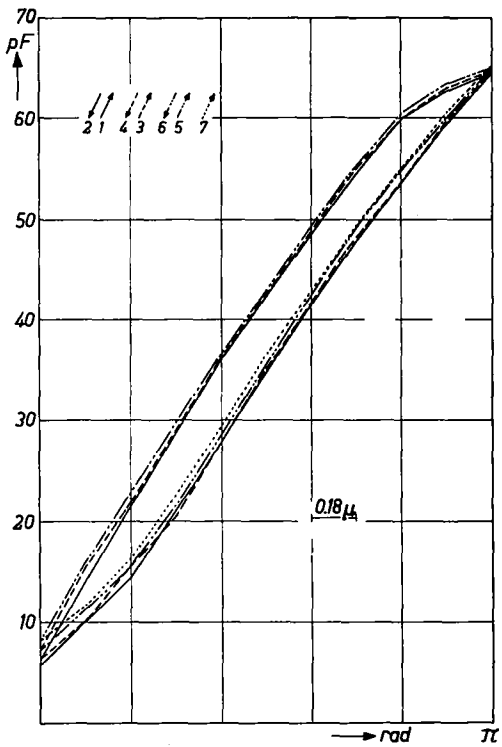


Fig. 3. Determination of the properties of the lever system. Vertical: capacity variations of the plates m condenser. Horizontal: angle of rotation of screws n (fig.2C)

The hysteresis behaviour of knife edge contacts points to a weakness in the experimental set up of the experiments by Dejaguin *et alii*. They retained a knife edge contact in their balance system on a place equivalent to e in our system.

Such a knife edge contact will favour a force-distance relationship of the plates m in which the force indicates a dependence on the distance which is too strong. This assumption is justified by the numerous zero values or very small values of the force at a fairly large distance where the deflection of the balance arm should already be small. The retaining force in the knife edge contact is in that case probably not zero. Thus, whilst for example a inverse

4th power relationship between forces and distance was found, it should be for example an inverse 2nd power relationship.

3.4 *Parallel position of the plates.* After placing two 6W lamps in the plane of the interface of the plates m, the distance of the lamps towards the plates being about 50 cm and the angle between the lines joining each lamp and the plates being 90°, then, judging by eye, the plates could fairly well be brought in parallel position. When the mean interplate distance was about 5 μ, the difference between the minimum and the maximum distance could be made $0.4\,\mu \pm 0.1\,\mu$. This was checked by bringing the plates together by simultaneous application of the levêr systems such, that the plates m touched. Touching was detected by a sudden decrease of the capacity of the condenser c. The moment of contact could accurately be determined, one of the reasons being that movements in vertical direction of the upper plates m were about 5 times magnified in similar movements of the upper plate c. The capacity measurement usually had to be made less sensitive in order to establish parallelism, the influence of vibrations being ruled out. The lever systems were used separately after the plates were touching, in order to find a good parallel position. Even after excercise there probably existed relative deflections of the planes of the plate surfaces of about 0.2μ, mainly owing to dust particles. Dust particles, may give a wrong impression of the relative position of the contacting plates and, moreover, experiments were preferably carried out as short as possible after a fairly parallel position was obtained, because dust particles may settle at any moment.

3.5 *Dust particles and cleaning procedures of the plates.* Dust particles larger than 2–3 μ were easily observed by projecting a 20–30 fold magnified image of the plates m on a screen.

When the interplate gap was larger than 3 μ, a white line was visible. Presence of dust particles was marked by interruptions of the white line. These interruptions were due to light diffraction and the dirt particles were often smaller than the distance between the plates as was shown by manipulating the lever systems and simultaneous observation of the capacity variations, if any, of the condenser c.

Dust particles were distributed at random on the plates. This is in agreement with results obtained by Williamson *et alii* [21]) in their work on the settling of dust particles on metal plates and is also in agreement with the distribution of dust particles on glass-or quartz plates as observed earlier. Complete removal of dust particles is difficult and was probably never effected. Rubbing with lens tissue or with chamois leather, both dipped in freshly distilled alcohol, was used as a cleaning procedure. It was important, that no removal of the plates m was necessary. Simple lifting of the beam k was sufficient for cleaning. An entirely different method was also used: a glass cell, investigated on the brass plate r was built around

the plates m. A strong current of freshly distilled, filtered alcohol was pumped through the gap between the plates m, where they were a few mm. apart and an alternating voltage (10–40 volt) was applied. The initial atmosphere in the system was dust free pure nitrogen gas. After alcohol was was pumped off and trapped by liquid air, the result, which depended on the quality of the alcohol, was generally the same as that obtained after rubbing.

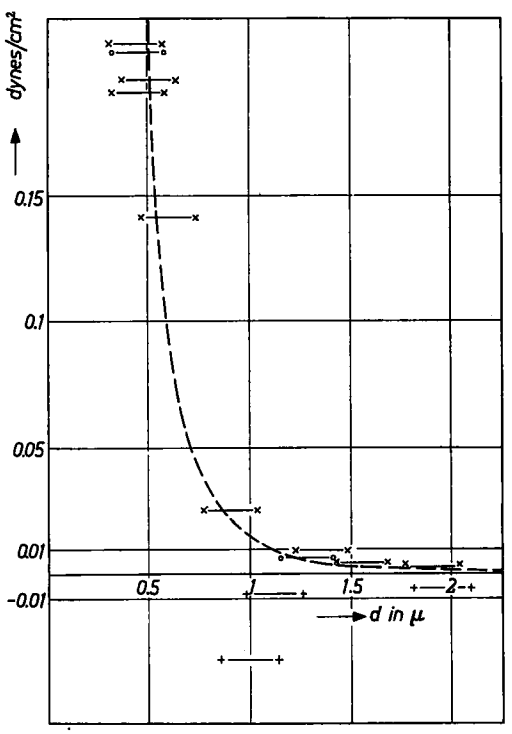


Fig. 4. Results ×——× chromium steel o——o chromium. Uncertainty of the determination of the distance between the plates is indicated by drawing horizontal lines instead of points. Any given measurement was often repeated. Some repulsions between aluminium plates are also given. (+——+). ------ Casimir's relation (eq. 1).

3.6 *Measurements of attractive forces and influence of obstacles between aluminium, glass and quartz plates.* Fig. 4 gives results for chromium steel, chromium and aluminium plates. Only effects that instantaneously occurred (i.e. in less than ½ minute) upon decreasing and increasing d were considered. The analysis of the results was done in the same way as described earlier [5]). It is seen, that the results of the first two of these metals do not seriously deviate from Casimir's predictions, though the attractions found are somewhat too large. Especially the dependence of the force upon the distance was marked. Whereas the attraction is strong at 0.3–0.5 μ, no effect at all could be detected at a distance of 2.5 μ. The plates at this distance were short circuited; no attraction resulted. However, the determination of the

mean distance was not precise. The lack of precision is symbolized by presenting the measurements as horizontal lines, not by points. It is easily seen, that a non parallel position of the plates m gives rise to an interpretation of the attractive force which has a too high force constant, when the mean distance is ultimately determined by a capacity measurement. No attractions could be measured between two aluminium plates. Upon bringing the plates together at a distance smaller than 2–3 μ, small repulsions were the only result. Some of these repulsions are also given in the figure. The origin of these repulsions is not clear, but the most probable explanation seems to be the existence of little sheets of aluminium oxide protruding from the surface after removal in air of dust particles. The behaviour of two aluminium plates brought close together resembles the behaviour of glass and quartz. In the latter case small repulsions were also observed, and large repulsions only, when a rigid dirt speck was present. Some information concerning the presence and the elastic and plastic properties of the obstacles between glass and quartz plates were gained in previous work [5]) of the author. In these earlier experiments the connection e and the balance beam k served as a spring, the parts at the left hand side of e, and the spring s being absent. The upper plate m could than be given an oscillation with a frequency of about 1 sec^{-1}. These oscillations could easily be followed by means of capacity variations or by observing oscillating interference fringes. When these oscillations were damped, obstacles were present between the plates. Observations of the way of damping gave information concerning the nature of the obstacles (the influence of the gas atmosphere was of course also taken into account). If it appeared that obstacles with a more or less elastic and plastic behaviour were present, they usually had a large influence upon the attractive measurements.

Thus, irrespective the force constant of the spring used (the force constant was varied over a factor 10 and preferably a spring with high force constant was used) the deflection of the upper plate m, when brought between 0.7 μ and 1.2 μ distance of the lower plate, was the same. The deflection was large, say 100Å, when the distance was 0.7 μ and was small, say 10Å when the distance was 1.2 μ. When the distance was made smaller than 0.7 μ, attractions were still often observed, but on the whole [11]) repulsions began to prevail. The value of the deflections given here are in qualitative agreement with the deflections, obtained by Kitchener and Prosser. Since these investigators worked with a spring which had a force constant, which was 10–100 times smaller, their attractive forces became systematically smaller in the same ratio.

It is only when obstacles are absent, that large deflections, inverse proportional to the force constant occur and that a small distance (not reported by Derjaguin *et alii* and by Kitchener and Prosser) can easily be obtained.

Oscillations can not be used to give information concerning obstacles if the force constant is small and consequently if the frequency of the oscillations is low, say 0.1 sec^{-1}. Then the influence of obstacles can not be distinguished from other effects, other (irregular) vibrations with an origin outside the system prevailing. This applied to the cases of Derjaguin and Kitchener and to our present apparatus and represents a weakness.

4. *Future work.* We have seen, that, though attractive forces, in qualitative agreement with Casimir's prediction are found, the following difficulties in our experimental set up are still present:

1) The potential differences between the plates can not be measured in our apparatus. Information concerning their value is useful however, as is explained above.

2) There are sometimes obstacles preventing attractive measurements. The experimental set up is now being modified such as to meet these two difficulties. Results will be given in a future publication.

Received 20-5-58.

REFERENCES

1) Casimir, H. B. G., Proc. kon. Ned. Akad. Wetenschap **51** (1948) 793.
2) Lifshitz, E. M., C.R. Acad. Sci. U.S.S.R. **97** no 4 (1954) 643.
3) Hamaker, H. C., Physica **4** (1937) 1058.
De Boer, J. H., Trans. Faraday Soc. **32** (1936) 21.
4) London, F., Z. physik. Chemie **B 11** (1931) 222.
5) Overbeek, J. Th. G. and Sparnaay, M. J., Discussions of the Faraday Soc. **18** (1954) 19.
6) Derjaguin, B. V. and Landau, L., Acta physicochim **14** (1941) 633; J. Expt, Theor. Physics (Russ.) **11** (1941) 802.
Verwey, E. J. W., Chem. Weekblad **39** (1942) 563.
Verwey, E. J. W. and Overbeek, J. Th. G., Theory of the Stability of Lyophobic Coiloids Elsevier 1948.
Kruyt, H. R., Colloid Science, Elsevier, vol. I, 1952.
7) Levich, V. G. and Derjaguin, B. V., Discussion of the Faraday Soc. **18**, (1954) 85; Bierman, A., J. Coll. Sci. **10** (1955) 231.
8) Von Buzagh, A., Kolloid Z. **47** (1929) 370; **51** (1930) 105.
9) Overbeek, J. Th. G. and Sparnaay, M. J., Discussions of the Faraday Soc. **18** (1954) 12.
10) Bonhoeffer, K. F. and Franck, U. F., Z. Electrochemie **55** (1951) 180.
11) Ref. 5, sections 1.4 and 1.9*a*.
12) Brüche, E. and Poppa, H., J. Soc. glass Techn. **40** (1956) 513.
13) See also Brunauer Physical Adsorption of gases and Vapours, Oxford 1944 on a 7 fold increase of the surface area of a glass surface after rinsing with acids (Chapter X).
14) Tolansky, S., Multiple beam interferometry, Clarendon Press Oxford 1948.
15) Sparnaay, M. J., to be published.
16) Derjaguin, B. V., Abrikossova, J. J. and Lifshitz, E. M., Quart. Rev. **10** (1956) 295.
Derjaguin, B. V. and Abrikossova, J. J., Discussion of the Faraday Soc. **18** (1954) 33.
17) Kitchener, J. A. and Prosser, A. P., Proc. roy. Soc. **A 242** (1957) 403.
18) Ref. 5 table 5.
19) Howe, P. G., Benton, D. F. and Puddington, J. E., Can J. Chem. **33** (1955) 1375.
20) Everett, D. H. and Whitton, W. J., Trans. Faraday Soc. **48** (1952) 749.
Everett, D. H. and Smith, F. W., Trans. Faraday Soc. **50** (1954) 187.
Everett, D. H., Trans. Faraday Soc. **50**, (1954) 1077; **51** (1955) 1551.
21) Williamson, B. P., Green wood, J. A. and Harris, J., Proc. roy. Soc. **A 237** (1956) 560.